

Project Description

Team: Flash

Problem Statement

What is the problem you would like to solve?

(Provide a detailed description of the problem your team is aiming to solve.)

In a manufacturing environment, process complexity often leads to inefficiencies such as production delays, increased machine downtime, and quality issues. As processes become more complex, the time and resources needed for each unit of production also rise, negatively affecting overall throughput. By optimizing the scheduling and sequencing of these complex processes, the goal is to minimize production losses, enhance efficiency, and improve the cost-effectiveness of the production line. This optimization is particularly crucial in industries like automotive or electronics manufacturing, where high process variability can significantly impact productivity.

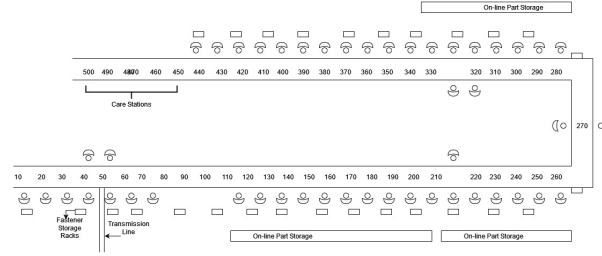


Figure 1: Demonstration diagram for the whole process.

Optimization Model

1. Decision Variables

$x[i, j]$ = Binary variable indicating whether task i is scheduled at time j ($x[i, j] \in \{0, 1\}$)

$y[k]$ = Binary variable indicating whether machine k is working or idle ($y[k] \in \{0, 1\}$)

$P[i]$ = Number of people required for task i (integer)

$Z[k]$ = Binary variable indicating whether person k is idle ($Z[k] \in \{0, 1\}$)

$Pr[i, k]$ = Binary variable indicating if process i is scheduled at station k ($Pr[i, k] \in \{0, 1\}$)

$w[i, k]$ = Binary variable indicating if process i should occur at station k ($w[i, k] \in \{0, 1\}$)

$T[i]$ = Completion time of task i (continuous)

$V[k]$ = Binary variable indicating if machine k requires a tool ($V[k] \in \{0, 1\}$)

$U[k]$ = Binary variable indicating if process at machine k is a final fitment ($U[k] \in \{0, 1\}$)

$T_{assembly}[i]$ = Time to assemble process i (continuous, varies per machine)

$Delay[i]$ = Delay time for process i (6 or 7 seconds depending on the task)

2. Constraints

$$T[i, k] \geq 6 + 1 \cdot V[k] \quad \forall i, k \quad (1)$$

$$P[i] \cdot (T[i] - T[i - 1]) \geq 25 \quad \forall i \quad (2)$$

$$T[90] = 0 \quad (3)$$

$$T[100] = 0 \quad (4)$$

$$T[40] \leq 36 \quad (5)$$

$$T[i] \leq 42 \quad \forall i \ (i \neq 40) \quad (6)$$

$$T[i] \geq T[i - 1] + \text{Delay}[i] \quad \forall i \quad (7)$$

$$T[i] \leq \text{Max Time}[i] \quad \forall i \quad (8)$$

$$x[i, j] + x[i', j] \leq 1 \quad \forall i \neq i', j \quad (9)$$

$$T[i] \geq \text{Complexity Factor}[i] \cdot \text{Base Time}[i] \quad \forall i \quad (10)$$

$$T_{\text{assembly}}[i] \geq 3 \quad \forall i \quad (11)$$

$$V[k] = 1 \quad \text{if a nut runner is needed at station } k \quad (12)$$

3. Objective function

$$\text{Minimize } Z = \sum_{i=1}^{45} C[i] \cdot T[i] + \sum_{k=1}^{45} \text{Idle Cost}[j, k] \cdot y[k] + \sum_{i=1}^{45} \text{Idle Cost of P}[i] \quad (13)$$

4. Parameters

$$\text{Cost of line breakdown} = 11,000 \text{ USD per hour} \quad (14)$$

$$\text{Cost of electricity} = 0.3 \text{ USD per unit (kWh)} \quad (15)$$

$$\text{Cost of compressed air} = 0.1 \text{ USD per cubic mm} \quad (16)$$

$$\text{Process completion time} = \text{Predefined for each process} \quad (17)$$

$$\text{Cost of manpower} = 10 \text{ USD per hour per man} \quad (18)$$

Analysis

What kind of analysis do you want to do?

0.1 Sensitivity Analysis

- **Station Idle Time Sensitivity:** Investigate how changes in the idle cost of machines and people affect the total cost.
- **Delay Sensitivity:** Analyze the impact of varying the delay times (6 or 7 seconds) on the overall schedule and total cost.
- **Constraint Sensitivity:** Explore how relaxing or tightening specific constraints (e.g., assembly times, task sequencing) influences the solution.

0.2 Feasibility and Optimization

- Ensure that the model remains feasible under all imposed constraints (e.g., no overlapping tasks, adherence to delay times, etc.).
- Use optimization algorithms such as Mixed-Integer Linear Programming (MILP) to minimize the objective function while satisfying all constraints.

0.3 Objective Function Sensitivity

- Study how changes in the costs of downtime, electricity, and compressed air influence the overall cost. This helps to identify the most critical cost drivers in the production line.

Data Requirements

Where can you find the data you need?

We generated some data according using some rules.

Next Steps

What are the next steps?

(Outline a list of next steps, assign responsibilities, and include deadlines if possible.)

1. Formulate the optimization problem.
2. Gather necessary data.
3. Start implementing the model in Julia.
4. Conduct preliminary analysis and testing.
5. Prepare for the in-class presentation on November 14.

Important Course Dates

- **November 12** – Due date for project description (either a filled-in version of this form or a draft of the final project Julia notebook template).
- **November 14** – In-class meeting with your group. Present the current status of your project to another group for feedback.
- **December 6** – In-class meeting with your group. Present the current status of your project to another group for feedback. Show finalized implementation and preliminary results.
- **December 13** – Full project report due.